Energy Procurement **Optimization**

Multi-Objective Linear Programming (MOLP) Approach

- **Decision Tree**
- **Linear Programming**
- **Green Energy**

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Executive Summary

Objective

A manufacturing plant must procure electricity every hour to meet production demand.

Determine the **optimal hourly portfolio mix** – how much electricity to procure from each source – to balance simultaneously:

- (1) **minimize total cost;**
- (2) **minimize supply risk**; and
- (3) **maximizing green energy**

Context & Challenges

No single source satisfies all three sub-objectives. The optimal mix depends on prices, demand levels, and how much the company values sustainability.

Three available sources:

	Cost	Supply Reliability	Green Score
(1) Standard Contract	Medium	Generator failure possible	Not certified
(2) RE Certified Contract	High	Generator failure possible	GO certified
(3) Spot Market	Low (avg)	Higher if reported timely to operator	Not certified

Proposed Model

Scenario-based MOLP with:

(1) **Decision tree** structures demand & price scenarios; and

(2) **Multi-objective Linear Programming** solves optimal portfolio allocation in expectation across all scenarios.

Pipeline

1 – Decision Framework
Define scenarios & MOLP problem structure

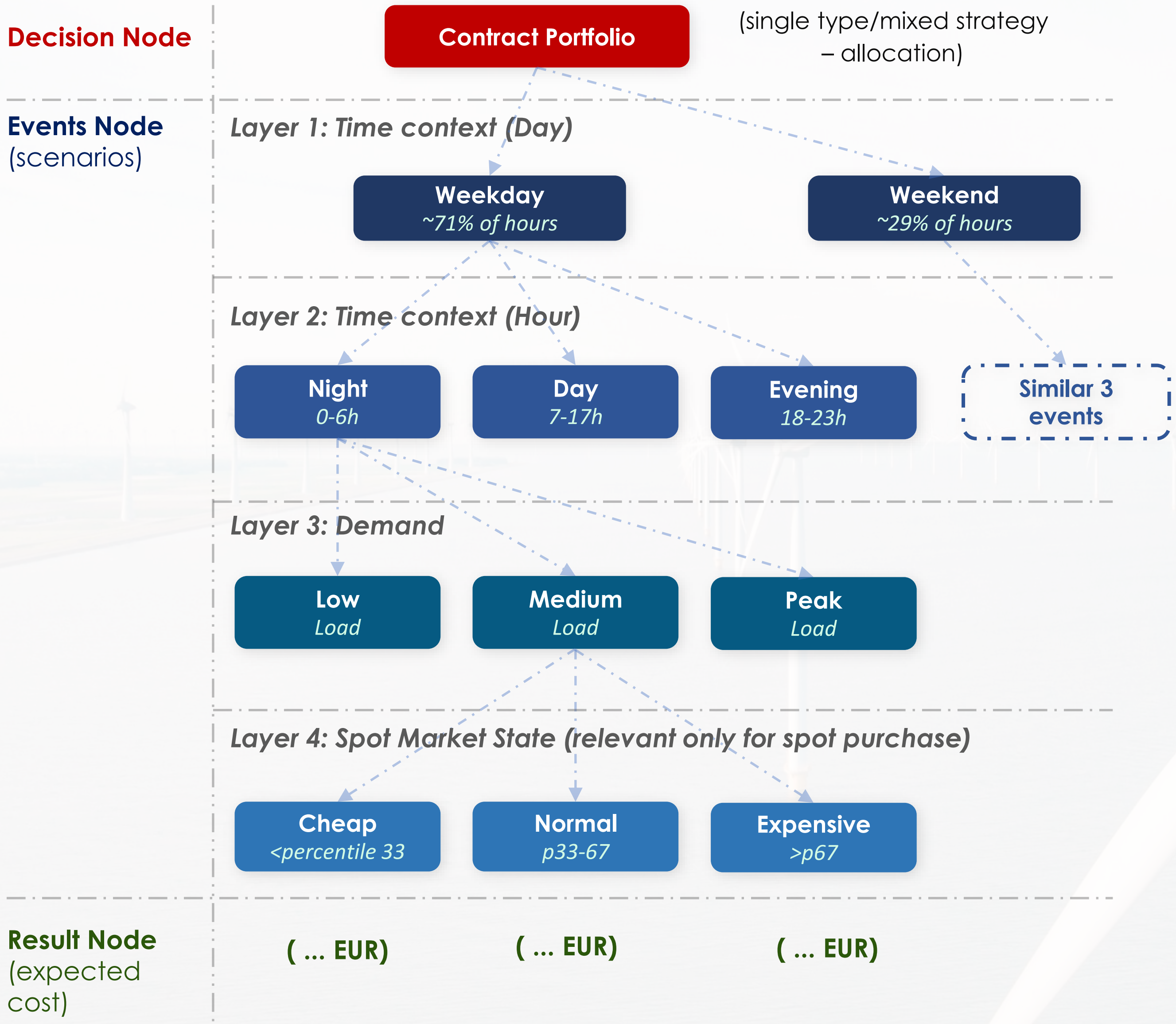
2 – Data Collection
Obtain firm and electricity market data.

3 – Scenario Building
Estimate probability for decision trees from data.

4 – Solve & Optimize
LP solver per scenario to obtain optimal portfolio.

5 – Sensitivity Analysis
Across different weights/contract premium.

Decision Tree



Scenario Structuring

- **Probabilities of each scenario are estimated from data:**
  - Demand tiers → Load_Type frequency in DAEWOO dataset
  - Price tiers → Percentile splits (P33 / P67) from ENTSO-E Finland spot prices
  - Time context → Weekday/weekend & hour distributions from data
- Each leaf node probability = product of conditional probabilities along its path.
- **Total scenarios:**
  - If spot prices involved: up to 54 leaf nodes (2 × 3 × 3 × 3)
  - Fixed contracts only: 1 leaf node

Optimization Model – Objectives

Aggregate Objective Function

$$\min_{x_1(t),\, x_2(t),\, x_3(t)} \sum_s p(s).\left[w_1.\,\mathrm{Cost}(s) + w_2.\,\mathrm{Supply\ Risk}(s) - w_3.\,\mathrm{Green\ Score}(s)\right]$$

$x_1(t)$ $x_2(t)$ $x_3(t)$ **Decision Variables (solved by LP)** – MWh bought from Standard Contract/ RE-Certified Contract/ Spot Market

w_1 w_2 w_3 **Priority Weights (set by the company)** – How much company values cost savings/ supply reliability/ green score.

s p(s) **scenario s** (combination of time context, demand level, spot price state) & **probability p(s)** of each scenario

① Cost $= \sum_t \sum_{i=1}^{3} p_i(t).\,x_i(t)$

where:

- p_1 standard contract price = E[spot] × 1.10
- p_2 RE contract price = E[spot] × 1.15
- $p_3(t)$ spot price at hour t

Multiply price [€/MWh] × quantity [MWh] → cost [€]

② Supply Risk $= \sum_t \mathrm{short\_fall}(t)\,.\,\mathrm{p\_bal}$

where:

short_fall (t) = max(0, $\epsilon.(x_1 + x_2)$): possible amount that contract generator fails to deliver

ϵ: fraction of committed amount contract generator fails to deliver

p_bal : price to buy from balancing market

Captures production loss from unmet demand

③ Green Score $= \sum_t x_2(t)$

i.e. total amount bought from RE-certified contracts with GO certificates

Unit: MWh of certified green procurement

Note: t = each hour in planning horizon (8,760 hrs/year)

Assume baseline **$\epsilon(t)$ = 0.02,**

Optimization Model – Programming

The **LP solves for $x_1(t)$, $x_2(t)$, $x_3(t)$** that **minimize the probability-weighted sum of all three objectives** across all defined scenarios.

Constraints:

	Inequality/Equation	Explanation
(1) Demand Coverage	$\mathbf{x_1(t) + x_2(t) + x_3(t) = D(t)}$	Must cover demand at each hour
(2) Standard Contract Cap	$\mathbf{x_1(t) \leq \overline{C_1}}$	Cannot exceed agreed contract capacity
(3) RE Contract Cap	$\mathbf{x_2(t) \leq \overline{C_2}}$	Cannot exceed agreed contract capacity
(4) Non-negativity	$\mathbf{x_1(t),\ x_2(t),\ x_3(t)\ \geq\ 0}$	Cannot procure negative quantities

* *Note: Assume* $\overline{C_1}, \overline{C_2} = \mathbf{50\%, 20\%}$ of average $\mathbf{D(t)}$

Expected Outputs

01

Scenario Probability Table

Cross-tabulated probabilities for all leaf nodes, derived from data.

02

Optimal Portfolio Mix

Hourly x_1 / x_2 / x_3 allocation per scenario.

03

Cost vs Baseline

Expected savings vs naive always-spot strategy, quantified in €/year.

04

Trade-off Frontier

How optimal cost and green score shift as w_1, w_2, w_3 weights change.

05

Contract Premium Sensitivity

Optimal mix at 5% / 10% / 15% premium – value of contract stability.

06

Supply Reliability Analysis

Impact of contract generators' failure to deliver on cost/shortfall